

ACADEMIC POSITIONS

Postdoctoral Researcher

10/2024 - Present

HCI and AI

Computational Behavior Lab, Aalto University, Finland.

Advisor: Prof. Antti Oulasvirta

- Developing Reinforcement Learning based approaches for modeling human decision making under partial observability.
 - Researching methods for cost-aware Bayesian optimization for design and prototyping.
 - Teaching courses (Master and Ph.D. level) and student supervision (interns and M.Sc. thesis).
-

EDUCATION

Ph.D.

10/2018 - 07/2024

Computer Science (HCI and AI)

Advanced Interactive Technologies, ETH Zürich, Switzerland.

Advisors: Prof. Otmar Hilliges & Prof. Christian Holz

Committee: Prof. Christian Holz, Prof. Antti Oulasvirta, and Prof. Roderick Murray-Smith

- Multi-agent reinforcement learning for user modeling and design optimization.
- Model-predictive control for haptic devices that incorporates predictive models of human behavior for agency-automation trade-off.
- Design optimization of novel actuators and sensing algorithms for haptic devices.
- Teaching (B.Sc., M.Sc., and Ph.D.) as well as student supervision.

Double-degree M.Sc.

09/2016 - 08/2018

Computer Science (Interactive Intelligent Systems)

Aalto University & Twente University.

- With Honors (Distinction) from Aalto University
- Focus on Machine Learning, Intelligent Systems, Multi-Agent Systems and Entrepreneurship

Exchange (Sophomore Year)

01/2015-06/2015

Industrial Design

Carnegie Mellon University, USA

- Product and computational design.

B.Sc.

09/2012 - 02/2016

Industrial Design

Eindhoven University of Technology, the Netherlands

PROFESSIONAL EXPERIENCE**Research Scientist Intern**

05/2023-10/2023

Explainable AI

Meta Reality Labs, Redmond, USA

- Bayesian user models for better explainable AI in the context of adaptive interfaces.
- Completed a full research project to successful publication ([J.1.]).

Graduate Intern

02/2018-10/2018

Human-Computer Interaction & Model Predictive Control

Advanced Interactive Technologies, ETH Zürich, Switzerland (Advisor: Prof. Otmar Hilliges)

- Invited visiting student for my Master thesis.
- Focused on Model Predict Control strategies for haptics.
- Built a custom haptic system.
- Resulted in a Conference Paper and Demo.

Summer Intern

05/2017-08/2017

Computational Design

User Interfaces, Aalto University, Finland (Advisor: Prof. Antti Oulasvirta)

- Implemented different metrics for automatic interface aesthetics.
- Resulted into the Aalto Interface Metric Server.

Intern

09/2014-01/2015

Industrial Design & Prototyping

Studio Sophisti, Amsterdam, the Netherlands

- Prototyping, Software, Electrical Engineering and Design for interactive devices.
- First iteration on a Core77 design award winner.
- Worked for Lego, Disney and Hasbro.

TEACHING**(Co)-Lecturer**

<i>Computational Design and Interaction (120, M.Sc. + PhD)</i>	2025
<i>Seminar on Human-AI Interaction (20, M.Sc. + PhD)</i>	2025
<i>Seminar on Human Performance Capture (20, M.Sc. + PhD)</i>	2024
<i>Seminar on Computational Haptics (20, M.Sc. + PhD)</i>	2020, 2021

Individual Lectures

<i>Human-Computer Interaction: Computational Rationality (300, B.Sc.)</i>	2023
<i>Human-Computer Interaction: (Computational) Haptics (100, B.Sc.)</i>	2021-2023
<i>Human-Computer Interaction: Combinatorial Optimization (100, B.Sc.)</i>	2021, 2022
<i>Human-Computer Interaction: Combinatorial Optimization (30, Industry)</i>	2020

Teaching Assistant

<i>Computer Science I (>600, B.Sc.)</i>	2022
<i>Ubiquitous Computing (20, B.Sc.)</i>	2020, 2021
<i>Seminar on Advanced topics in Technical HCI (20, M.Sc. + PhD)</i>	2020, 2021

<i>Human-Computer Interaction (100-300, B.Sc.)</i>	2020-2023
<i>Seminar in Computational Interaction (20, M.Sc. + PhD)</i>	2019
<i>Fairness, Equality and Accountability in Machine Learning (25, M.Sc.)</i>	2019

COMMUNITY SERVICE & ORGANIZING COMMITTEES

<i>Associate Chair</i>	2025
Conference on Human Factors in Computing Systems (CHI)	
Conference on Intelligent User Interface (IUI)	

<i>Co-Chair, Data</i>	2021, 2022
Symposium on User Interfaces, Systems and Technologies (UIST)	
- Combine the different systems a conference uses and manage all data streams. - Collaborate with different chairs.	

<i>Co-Chair, Virtual Experiences and Operations</i>	2020
Symposium on User Interfaces, Systems and Technologies (UIST)	
- Transitioning a physical conference into virtual-only. - Investigate the needs and solutions for virtual conferences. - Manage all data streams.	

REVIEWING

I am a distinguished reviewer for TOCHI. I review for all HCI conferences, including CHI, UIST, IUI, EICS and NordiCHI. I also review for journals such as Transactions on Haptics, and IEEE Sensors. Furthermore, I review for ML conferences (e.g., ICLR). Occasionally, I review for robotics such as HRI.

INVITED TALKS & PANELS

MBZUAI, AI Research Talk	2026
Creating Machines that Better Understand and Work with People	
Summer School on Computational Interaction	2026
Panel on Active Inference	
Summer School on Computational Interaction	2026
Reinforcement Learning for HCI	
UbiComp-ISWC	2025
Doctoral Consortium Panel	
Sorbonne University	2025
Planning with Human Decision Models in Intelligent Systems	
Summer School on Computational Interaction	2025
Panel on Computational Interaction	

Summer School on Computational Interaction Reinforcement Learning for HCI	2025
UCL User Agency and System Automation in Interactive Intelligent Systems	2024
TU Delft User Agency and System Automation in Interactive Intelligent Systems	2024
TU Eindhoven User Agency and System Automation in Interactive Intelligent Systems	2024
CMU, HCII User Agency and System Automation in Interactive Intelligent Systems	2023
CMU, Augmented Perception Lab Multi-Agent Reinforcement Learning for Adaptive UIs	2022

AWARDS

<i>UIST Super Hero Award</i>	2020
For going above and beyond to transfer the conference from physical to a virtual venue.	
<i>Honorable Mention - World Haptics Conference</i>	2020
<i>NASA Europa Challenge Finalist</i>	2019
<i>EIT Digital Excellence Scholarship</i>	2019

Grants

<i>SIGCHI Diversity Grant (5000 Euros)</i>	2020
To enable people who otherwise cannot attend UIST to join.	

STUDENT SUPERVISION

Antti Immonen , Master Thesis Instance-level Contrastive Learning Methods for User Interfaces	2025
Renate Zhang , Summer Internship Bayesian Optimization for Prototyping Interactive Devices Published in <i>CHI 2026</i>	2025
Morgan Wang , Research Assistant Bayesian Optimization for Prototyping Interactive Devices Published in <i>CHI 2026</i>	2025
Yugdeep Bangar , Master Thesis User Interface Optimization for the Quantified Self Co-supervised with Prof. Alan Hanjalic, TU Delft	2024
Caroline Sauget , Master Thesis Deep Reinforcement Learning for Sustainability	2022

Mengfan Wu, Master Thesis 2021
Electromagnetic Tracking via Deep Learning
Published in *IEEE Transactions on Magnetics*

Aline Abler, Master Thesis 2021
Building a Hedgehog Pin Array Haptic Interface
Published in *World Haptics Conference 2021*
Honorable Mention for Best Paper

References

Prof. Antti Oulasvirta, *Aalto University*, antti.oulasvirta@aalto.fi
Prof. Christian Holz, *ETH Zürich*, christian.holz@ethz.ch
Prof. Per Ola Kristensson, *University of Cambridge*, pok21@cam.ac.uk

PUBLICATIONS

Under Review & Ongoing

Only papers that are under review or of which I am lead researcher are listed.

- [WIP.1]: *Design Optimization with Multiple Evaluation Sources using Multi-Surrogate Bayesian Optimization*
Francisco Fernandes, **Thomas Langerak**, Mira Kernen, Danqing Shi, Ardak Alipova, and Antti Oulasvirta. Accepted with Revisions.
- [WIP.2]: *EvoBO: Design Spaces that Evolve with Design in Bayesian Optimization*
Thomas Langerak, Lena Hegeman, Yichi Liao, and Antti Oulasvirta. Under Review
- [WIP.3]: *RILe: Reinforced Imitation Learning*.
Mert Albaba, Sammy Christen, Christoph Gebhardt, **Thomas Langerak**, Michael J. Black, and Otmar Hilliges. Under Review.
- [WIP.4]: *Inverting POMDPs with Unknown Observation Functions and Rewards*. In Progress.
- [WIP.5]: *Learned Latent Rational Control for Modeling Human Gaze Behavior*. In Progress.

Conference Papers

- [C.1]: *Bayesian Optimization for Prototyping Interactive Devices*
Thomas Langerak, Renate Zhang, Morgan Wang, Per Ola Kristensson and Antti Oulasvirta. 2026. Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems.
Honourable Mention Best Paper
- [C.2]: *MARLUI: Multi-Agent Reinforcement Learning for Goal-Agnostic Adaptive UIs*.
Thomas Langerak, Sammy Christen, Mert Albaba, Christoph Gebhardt and Otmar Hilliges. 2024. Proc. ACM Hum.- Comput. Interact. 8.
- [C.3]: *Hedgehog: Handheld Spherical Pin Array based on a Central Electromagnetic Actuator*.
Aline Abler, Juan Zarate, **Thomas Langerak**, Velko Vechev and Otmar Hilliges. 2021. In World Haptics Conference.
Honourable Mention Best Paper
- [C.4]: *Omni: Volumetric Sensing and Actuation of Passive Magnetic Tools for Dynamic Haptic Feedback*.
Thomas Langerak*, Juan Zarate*, David Lindlbauer, Christian Holz, and Otmar Hilliges. 2020. In Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology.
- [C.5]: *Optimal Control for Electromagnetic Haptic Guidance Systems*.
Thomas Langerak, Juan Zarate, Velko Vechev, David Lindlbauer, Daniele Panozzo, and Otmar Hilliges. 2020. In Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology.
- [C.6]: *Contact-free Nonplanar Haptics with a Spherical Electromagnet*.
Juan Zarate*, **Thomas Langerak***, Bernhard Thomaszewski and Otmar Hilliges. 2020. IEEE Haptics Symposium.

Journal Papers

- [J.1]: *XAIUI: Mental-Model Aware Explainable AI for Adaptive Interfaces*.
Thomas Langerak, Kashyap Todi, Ben Lafreniere, Ruta Desai, and Tanya Jonker. 2025. Transactions on Interactive Intelligent Systems.
- [J.2]: *Robust Real-Time Tracking of Axis-Symmetric Magnets via Neural Networks*.
Mengfan Wu*, **Thomas Langerak***, Juan Zarate and Otmar Hilliges. 2024. IEEE Transactions on Magnetism.

Auxiliary: Demos, Posters, and Workshops

- [A.1]: *RILe: Reinforced Imitation Learning*.
Mert Albaba, Sammy Christen, Christoph Gebhardt, **Thomas Langerak**, Michael J. Black, and Otmar Hilliges. 2024. 7th Robot Learning Workshop: Towards Robots with Human-Level Abilities (ICLR)
- [A.2]: *Generalizing User Models through Hybrid Hierarchical Control*.
Thomas Langerak, Sammy Christen, Anna Feit and Otmar Hilliges. 2021. In Reinforcement Learning for Humans, Computer, and Interaction (CHI 2021 Workshop).
- [A.3]: *A Demonstration on Dynamic Drawing Guidance via Electromagnetic Haptic Feedback*.
Thomas Langerak, Juan Zarate, Velko Vechev, Daniele Panozzo, and Otmar Hilliges. 2019. In The Adjunct Publication of the 32nd Annual ACM Symposium on User Interface Software and Technology.
- [A.4]: *Aalto Interface Metrics (AIM): A Service and Codebase for Computational GUI Evaluation*.
Antti Oulasvirta, Samuli De Pascale, Janin Koch, **Thomas Langerak**, et al. 2018. In The 31st Annual ACM Symposium on User Interface Software and Technology Adjunct Proceedings.

Dissertation

- [D.1]: User Agency and System Automation in Interactive Intelligent Systems

Patents

- [P.1]: Mental Model Aware Explainable Artificial Intelligence For Intelligent User Interfaces US Patent 20250123860